Taller 3

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1. Grafique las curvas de los seno de Taylor de varios ordenes para los siguientes casos

a. $f(x) = \ln(x)$

Mínimo de dos a 5to orden

con 1

$$f(x) = \ln(x)$$

$$f(x)' = \frac{1}{x}$$

$$f(x)'' = -x^{-2}$$

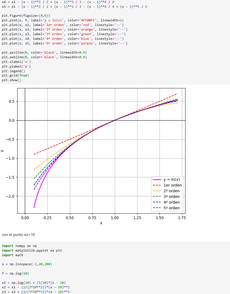
$$f(x)''' = 2x^{-3}$$

$$f(x)'''' = -6x^{-4}$$

$$f(x)''''' = 24x^{-5}$$

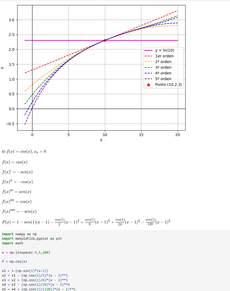
$$P(x) = \ln(1) + (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{6}(x-1)^3 - \frac{1}{12}(x-1)^4 + \frac{1}{20}(x-1)^5$$

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(0.1, 1.7, 200)

f = np.log(x)

x1 = (x - 1)
x2 = (x - 1)**2 / 2
x3 = x1 - (x - 1)**3 / 6
x4 = x1 - (x - 1)**2 / 2 + (x - 1)**3 / 3
x5 = x1 - (x - 1)**2 / 2 + (x - 1)**3 / 3 - (x - 1)**4 / 4 + (x - 1)**5 / 5

plt.figure(figsize=(6,4))
plt.plot(x, f, label='y = ln(x)', color='#FF0000', linewidth=2)
plt.plot(x, x1, label='1er orden', color='red', linestyle='--')
plt.plot(x, x2, label='2 orden', color='orange', linestyle='--')
plt.plot(x, x3, label='3er orden', color='green', linestyle='--')
plt.plot(x, x4, label='4T orden', color='blue', linestyle='--')
plt.plot(x, x5, label='5T orden', color='purple', linestyle='--')

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



con el punto en-10

In [2]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(-1,20,100)

f = np.log(10)

x1 = np.log(10) + (x-10)/10
x2 = x1 - (x-10)**2/(2*10)
x3 = x2 + (x-10)**3/(3*10)
x4 = x3 - (x-10)**4/(4*10)
x5 = x4 + (x-10)**5/(5*10)

plt.figure(figsize=(6,4))
plt.plot(x, f * np.ones_like(x), label='y = ln(10)', color='#FF0000', linewidth=2)
plt.plot(x, x1, label='1er orden', color='red', linestyle='--')
plt.plot(x, x2, label='2T orden', color='orange', linestyle='--')
plt.plot(x, x3, label='3T orden', color='green', linestyle='--')
plt.plot(x, x4, label='4T orden', color='blue', linestyle='--')
plt.plot(x, x5, label='5T orden', color='purple', linestyle='--')
plt.scatter([10, 2.3], [10, 0]*0, label='Punto (10,2.3)')

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



b. $f(x) = \cos(x), x_0 = 0$

$$f(x) = \cos(x)$$

$$f(x)' = -\sin(x)$$

$$f(x)'' = -\cos(x)$$

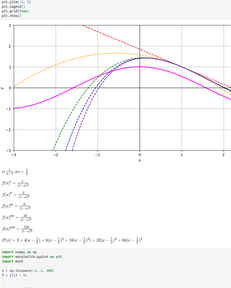
$$f(x)''' = \sin(x)$$

$$f(x)'''' = \cos(x)$$

$$f(x)''''' = -\sin(x)$$

$$P(x) = 1 - \sin(1)(x-1) - \frac{\cos(1)}{2}(x-1)^2 + \frac{\sin(1)}{6}(x-1)^3 + \frac{\cos(1)}{24}(x-1)^4 - \frac{\sin(1)}{120}(x-1)^5$$

In [3]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(-5,5,200)

f = np.cos(x)

x1 = (np.sin(1)*(x-1))
x2 = (np.cos(1)*(x-1)**2)/2
x3 = (np.sin(1)*(x-1)**3)/6
x4 = (np.cos(1)*(x-1)**4)/24
x5 = (np.sin(1)*(x-1)**5)/120

plt.figure(figsize=(6,4))
plt.plot(x, f, label='y=cos(x)', color='#FF0000', linewidth=2)
plt.plot(x, x1, label='1er orden', color='red', linestyle='--')
plt.plot(x, x2, label='2T orden', color='orange', linestyle='--')
plt.plot(x, x3, label='3T orden', color='green', linestyle='--')
plt.plot(x, x4, label='4T orden', color='blue', linestyle='--')
plt.plot(x, x5, label='5T orden', color='purple', linestyle='--')

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```



c. $f(x) = \sin x$

$$f(x) = \sin x$$

$$f(x)' = \frac{1}{1+x^2}$$

$$f(x)'' = \frac{-2x}{(1+x^2)^2}$$

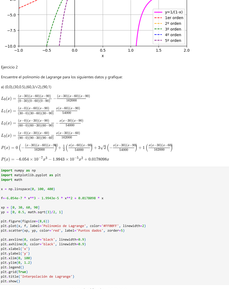
$$f(x)''' = \frac{-2}{(1+x^2)^2} + \frac{8x^2}{(1+x^2)^3}$$

$$f(x)'''' = \frac{16x}{(1+x^2)^3} - \frac{48x^3}{(1+x^2)^4}$$

$$P(x) = 0 + 0(x-\frac{1}{2}) + 0(x-\frac{1}{2})^2 + 16(x-\frac{1}{2})^3 + 32(x-\frac{1}{2})^4 + 64(x-\frac{1}{2})^5$$

In [4]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(-1, 2, 400)

f = 1/(1 - x)

x1 = x+1/2 x/2
x2 = x1 + (x)^2/2
x3 = x2 + (x)^3/6 - 1/12*x
x4 = x3 + (x)^4/24
x5 = x4 + (x)^5/120 - 1/192*x

plt.figure(figsize=(6,4))
plt.plot(x, f, label='y=1/(1-x)', color='#FF0000', linewidth=2)
plt.plot(x, x1, label='1er orden', color='red', linestyle='--')
plt.plot(x, x2, label='2T orden', color='orange', linestyle='--')
plt.plot(x, x3, label='3T orden', color='green', linestyle='--')
plt.plot(x, x4, label='4T orden', color='blue', linestyle='--')
plt.plot(x, x5, label='5T orden', color='purple', linestyle='--')

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```

```
C:\Users\emmor\AppData\Local\Temp\ipykernel_21644\2782575788.py:5: RuntimeWarning: divide by zero encountered in divide
  f = 1/(1 - x)
```



Ejercicio 2

Encuentre el polinomio de Lagrange para los siguientes datos y grafique:

a) (0,0),(30,0.5),(60,0.5),(90,0)

$$L_0(x) = \frac{(x-30)(x-60)(x-90)}{(0-30)(0-60)(0-90)} = \frac{(x-30)(x-60)(x-90)}{-162000}$$

$$L_1(x) = \frac{(x-0)(x-60)(x-90)}{(30-0)(30-60)(30-90)} = \frac{x(x-60)(x-90)}{54000}$$

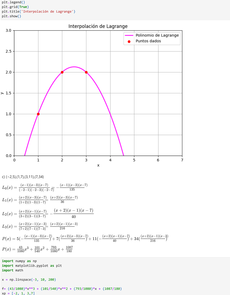
$$L_2(x) = \frac{(x-0)(x-30)(x-90)}{(60-0)(60-30)(60-90)} = \frac{x(x-30)(x-90)}{-54000}$$

$$L_3(x) = \frac{(x-0)(x-30)(x-60)}{(90-0)(90-30)(90-60)} = \frac{x(x-30)(x-60)}{162000}$$

$$P(x) = 0\left(\frac{(x-30)(x-60)(x-90)}{-162000}\right) + \frac{1}{2}\left(\frac{x(x-60)(x-90)}{54000}\right) + 0\left(\frac{x(x-30)(x-90)}{-54000}\right) + 0\left(\frac{x(x-30)(x-60)}{162000}\right)$$

$$P(x) = -8.054 \times 10^{-3}x^3 - 1.8943 \times 10^{-3}x^2 + 0.0176056x$$

In [5]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(0, 100, 400)

f=-8.054e-3 * x**3 - 1.9943e-3 * x**2 + 0.0176056 * x

y0 = [0, 30, 60, 90]
y1 = [0, 0.5, math.sqrt(2)/2, 1]

plt.figure(figsize=(6,4))
plt.plot(x, f, label='Polinomio de Lagrange', color='#FF0000', linewidth=2)
plt.scatter(y0, y1, color='red', label='Puntos dados', zorder=5)

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True, 0.2)
plt.legend()
plt.grid(True)
plt.title('Interpolación de Lagrange')
plt.show()
```



b) 0,1,2,3,4,5

$$L_0(x) = \frac{(x-1)(x-2)(x-3)(x-4)(x-5)}{(0-1)(0-2)(0-3)(0-4)(0-5)} = \frac{(x-1)(x-2)(x-3)(x-4)(x-5)}{-120}$$

$$L_1(x) = \frac{(x-0)(x-2)(x-3)(x-4)(x-5)}{(1-0)(1-2)(1-3)(1-4)(1-5)} = \frac{x(x-2)(x-3)(x-4)(x-5)}{24}$$

$$L_2(x) = \frac{(x-0)(x-1)(x-3)(x-4)(x-5)}{(2-0)(2-1)(2-3)(2-4)(2-5)} = \frac{x(x-1)(x-3)(x-4)(x-5)}{-12}$$

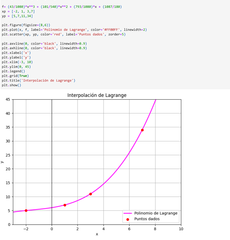
$$L_3(x) = \frac{(x-0)(x-1)(x-2)(x-4)(x-5)}{(3-0)(3-1)(3-2)(3-4)(3-5)} = \frac{x(x-1)(x-2)(x-4)(x-5)}{12}$$

$$L_4(x) = \frac{(x-0)(x-1)(x-2)(x-3)(x-5)}{(4-0)(4-1)(4-2)(4-3)(4-5)} = \frac{x(x-1)(x-2)(x-3)(x-5)}{-24}$$

$$L_5(x) = \frac{(x-0)(x-1)(x-2)(x-3)(x-4)}{(5-0)(5-1)(5-2)(5-3)(5-4)} = \frac{x(x-1)(x-2)(x-3)(x-4)}{120}$$

$$P(x) = 0\left(\frac{(x-1)(x-2)(x-3)(x-4)(x-5)}{-120}\right) + 1\left(\frac{x(x-2)(x-3)(x-4)(x-5)}{24}\right) + 1\left(\frac{x(x-1)(x-3)(x-4)(x-5)}{-12}\right) + 1\left(\frac{x(x-1)(x-2)(x-4)(x-5)}{12}\right) + 1\left(\frac{x(x-1)(x-2)(x-3)(x-5)}{-24}\right) + 0\left(\frac{x(x-1)(x-2)(x-3)(x-4)}{120}\right)$$

In [6]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(-1, 10, 200)

f1 =(x)(x-1)(x-3)(x-4)(x-5)/-120+(x)(x-2)(x-3)(x-4)(x-5)/24+(x)(x-1)(x-3)(x-4)(x-5)/-12+(x)(x-1)(x-2)(x-4)(x-5)/12+(x)(x-1)(x-2)(x-3)(x-5)/-24+(x)(x-1)(x-2)(x-3)(x-4)/120

y0 = [0, 1, 2, 3, 4]
y1 = [0,1,1,1,1]

plt.figure(figsize=(6,4))
plt.plot(x, f1, label='Polinomio de Lagrange', color='#FF0000', linewidth=2)
plt.scatter(y0, y1, color='red', label='Puntos dados', zorder=5)

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True, 0.2)
plt.legend()
plt.grid(True)
plt.title('Interpolación de Lagrange')
plt.show()
```



c) 1,2,3,5,7,11,17,19

$$L_0(x) = \frac{(x-2)(x-3)(x-5)(x-7)(x-11)(x-17)(x-19)}{(1-2)(1-3)(1-5)(1-7)(1-11)(1-17)(1-19)} = \frac{(x-2)(x-3)(x-5)(x-7)(x-11)(x-17)(x-19)}{-10}$$

$$L_1(x) = \frac{(x-1)(x-3)(x-5)(x-7)(x-11)(x-17)(x-19)}{(2-1)(2-3)(2-5)(2-7)(2-11)(2-17)(2-19)} = \frac{(x-1)(x-3)(x-5)(x-7)(x-11)(x-17)(x-19)}{15}$$

$$L_2(x) = \frac{(x-1)(x-2)(x-5)(x-7)(x-11)(x-17)(x-19)}{(3-1)(3-2)(3-5)(3-7)(3-11)(3-17)(3-19)} = \frac{(x-1)(x-2)(x-5)(x-7)(x-11)(x-17)(x-19)}{-16}$$

$$L_3(x) = \frac{(x-1)(x-2)(x-3)(x-7)(x-11)(x-17)(x-19)}{(5-1)(5-2)(5-3)(5-7)(5-11)(5-17)(5-19)} = \frac{(x-1)(x-2)(x-3)(x-7)(x-11)(x-17)(x-19)}{-24}$$

$$L_4(x) = \frac{(x-1)(x-2)(x-3)(x-5)(x-11)(x-17)(x-19)}{(7-1)(7-2)(7-3)(7-5)(7-11)(7-17)(7-19)} = \frac{(x-1)(x-2)(x-3)(x-5)(x-11)(x-17)(x-19)}{-24}$$

$$L_5(x) = \frac{(x-1)(x-2)(x-3)(x-5)(x-7)(x-17)(x-19)}{(11-1)(11-2)(11-3)(11-5)(11-7)(11-17)(11-19)} = \frac{(x-1)(x-2)(x-3)(x-5)(x-7)(x-17)(x-19)}{96}$$

$$L_6(x) = \frac{(x-1)(x-2)(x-3)(x-5)(x-7)(x-11)(x-19)}{(17-1)(17-2)(17-3)(17-5)(17-7)(17-11)(17-19)} = \frac{(x-1)(x-2)(x-3)(x-5)(x-7)(x-11)(x-19)}{-160}$$

$$L_7(x) = \frac{(x-1)(x-2)(x-3)(x-5)(x-7)(x-11)(x-17)}{(19-1)(19-2)(19-3)(19-5)(19-7)(19-11)(19-17)} = \frac{(x-1)(x-2)(x-3)(x-5)(x-7)(x-11)(x-17)}{144}$$

In [7]:
```
import numpy as np
import matplotlib.pyplot as plt
import math

x = np.linspace(-5,20,200)

f1 =(x)(x-3)(x-5)(x-7)(x-11)(x-17)(x-19)/-10+(x)(x-1)(x-5)(x-7)(x-11)(x-17)(x-19)/15+(x)(x-1)(x-2)(x-5)(x-7)(x-11)(x-17)(x-19)/-16+(x)(x-1)(x-2)(x-3)(x-7)(x-11)(x-17)(x-19)/-24+(x)(x-1)(x-2)(x-3)(x-5)(x-11)(x-17)(x-19)/-24+(x)(x-1)(x-2)(x-3)(x-5)(x-7)(x-17)(x-19)/96+(x)(x-1)(x-2)(x-3)(x-5)(x-7)(x-11)(x-19)/-160+(x)(x-1)(x-2)(x-3)(x-5)(x-7)(x-11)(x-17)/144

y0 = [1, 2, 3, 5, 7, 11, 17, 19]
y1 = [1,1,1,1,1,1,1,1]

plt.figure(figsize=(6,4))
plt.plot(x, f1, label='Polinomio de Lagrange', color='#FF0000', linewidth=2)
plt.scatter(y0, y1, color='red', label='Puntos dados', zorder=5)

plt.axvline(0, color='black', linewidth=0.5)
plt.axhline(0, color='black', linewidth=0.5)
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True, 0.2)
plt.legend()
plt.grid(True)
plt.title('Interpolación de Lagrange')
plt.show()
```